

A PROJECT TITLE

ON

Crop Recommendation Using Multinomial Logistic Regression

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Name of Title: Crop Recommendation Using Multinomial Logistic Regression

1. Abstract:

Agriculture plays a vital role in ensuring food security and supporting the economy. Selecting the most suitable crop for cultivation is a challenging task due to varying soil properties, climatic conditions, and environmental factors. Incorrect crop selection can result in reduced crop yield, financial losses, and inefficient utilization of agricultural resources. Traditional crop recommendation methods mainly depend on farmers' experience and manual analysis, which may not always provide accurate recommendations. Machine learning techniques provide an efficient solution by analyzing agricultural data and identifying patterns that help recommend the most suitable crop. This project proposes a machine learning-based crop recommendation system using the Multinomial Logistic Regression algorithm. The system uses agricultural parameters such as nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, pH value, and rainfall to recommend the most appropriate crop from multiple crop categories. The dataset undergoes preprocessing steps including data cleaning, handling missing values, encoding categorical variables, feature scaling where required, and splitting the dataset into training and testing sets. The Multinomial Logistic Regression model predicts the probability of each crop category based on the given soil and environmental conditions and recommends the crop with the highest probability. The performance of the model is evaluated using metrics such as Accuracy, Precision, Recall, F1-Score, and Confusion Matrix. The proposed system aims to assist farmers in selecting the most suitable crop for cultivation, thereby improving agricultural productivity, maximizing crop yield, and reducing the risk of crop failure. This project demonstrates the effectiveness of machine learning in solving real-world agricultural problems through data-driven decision-making. Future enhancements include integrating real-time weather data, IoT-based soil sensors, satellite imagery, and mobile applications to provide dynamic and location-specific crop recommendations.

2. INTRODUCTION:

Agriculture is one of the most important sectors that contributes significantly to the economy and food production. The success of farming mainly depends on selecting the right crop according to soil nutrients and climatic conditions. Farmers often face difficulties in choosing suitable crops because of changing weather patterns, varying soil fertility, and limited access to scientific decision-making tools. Selecting an unsuitable crop can reduce productivity, increase cultivation costs, and result in financial losses. Traditional crop recommendation methods mainly rely on farmers' knowledge and experience. Although these methods have been practiced for many years, they may not always provide accurate recommendations under changing environmental conditions. Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML) have enabled the development of intelligent systems that can analyze agricultural data and provide accurate crop recommendations.

This project focuses on recommending suitable crops using the Multinomial Logistic Regression algorithm. The algorithm is capable of classifying data into multiple crop categories based on various agricultural parameters such as nitrogen, phosphorus, potassium, temperature, humidity, pH value, and rainfall. By analyzing these features, the model predicts the most appropriate crop for a given set of environmental conditions. The project follows a systematic machine learning process involving data collection, preprocessing, feature engineering, model training, testing, evaluation, and prediction. The performance of the model is evaluated using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix to measure its effectiveness in recommending crops. The proposed system can support farmers by providing reliable crop recommendations that improve agricultural productivity and optimize resource utilization. It can also assist agricultural organizations and researchers in making data-driven decisions for sustainable farming. This project demonstrates how machine learning can contribute to modern agriculture by improving crop selection and supporting smart farming practices.

3. SOFTWARE REQUIREMENTS:

Software	Specification
Operating System	Windows 10/11, Linux, or macOS
Programming Language	Python 3.10 or above
IDE	Jupyter Notebook, VS Code, or PyCharm
Machine Learning Library	Scikit-learn
Data Processing	Pandas, NumPy
Data Visualization	Matplotlib, Seaborn
Classification Algorithm	Multinomial Logistic Regression
Model Evaluation	Accuracy, Precision, Recall, F1-Score, Confusion Matrix
Dataset Source	Kaggle Crop Recommendation Dataset
Version Control	Git a??


Signature of the Guide


Course Instructor